

Rohan Sonthalia

sonthaliarohan@gmail.com | +91 8334977754 | [linkedin.com/in/rohan-sonthalia/](https://www.linkedin.com/in/rohan-sonthalia/) | github.com/rohan-sonthalia | Bengaluru, India

EDUCATION

University of Wisconsin-Madison | *Visiting International Student, Computer Science* | GPA: 3.91/4.00 | Sept 2023 - Dec 2023
SRM Institute of Science and Technology | *Bachelor of Technology, Computer Science* | GPA: 9.76/10.0 | May 2025
Don Bosco School Park Circus | *High School Diploma* | Percentage: 94.5% | Mar 2021

EXPERIENCES

Associate AI/ML Engineer | *Optum (UHG)* | Bengaluru, India | July 2025 - Present

- Inventor on 3 patents filed and approved internally (USPTO provisional filing in preparation); won the AI Platform Hackathon and the Provider Services Organisation hackathon; delivered a technical update to 1,200+ employees at the OptumInsight Town Hall.
- Designed a parameter efficient dual-SLM architecture with adapter hot swaps for generating claim settlement summaries, reducing LLM token costs by over 90%.
- Built a Contract Intelligence platform using document parsing and LLM-based extraction pipelines to digitize and surface structured insights from contracts across multiple business lines.
- Developed a multi-agent chatbot for OptumRx's claims portals using LangGraph, implementing agent-to-agent handoffs and dynamic context windowing to handle complex member queries end-to-end.
- Led end-to-end automation of the claim appeals process — partnered with stakeholders and analysts to decompose workflows into AI agent modules, cutting manual processing time 75%+ and delivering \$10M+ in annual savings.
- Automated root cause analysis and data standardization for denied claims by applying LLM-based semantic column matching.
- Cut daily claims-summarization processing time from 33 hours to just over 3 hours (>30K claims, 4-hour SLA) by implementing a queuing mechanism and Horizontal Pod Autoscaling.

MITACS Globalink Research Intern | *Dumeaux Lab @ University of Western Ontario* | Ontario, Canada | Jun 2024 - Aug 2024

- Performed Deep Archetypal Analysis on gut microbiome data to identify non-linear combinations of Gene Expression Programs.
- Compared results of multiple deepAA models on compositional data, preprocessed through log-ratio transformations.

Software Engineer Intern | *Tune AI (formerly NimbleBox.ai)* | Chennai, India | Jan 2024 - May 2024

- Curated Automated API Testing suites for web servers using Golang.
- Implemented Structured Language Generation for improving output and control of Large Language Models.

Research Assistant | *Solis-Lemus Lab @ Wisconsin Institute of Discovery* | University of Wisconsin, Madison | Jul 2023 - Sept 2024

- Implemented an extension of the BioKlustering project that aims to draw phenotype conclusions from genomic data sequences.
- Leveraged transformers for embedding DNA sequence in the form on FASTA files, into vectors.
- Created a neural network to draw predictions from nonlinear genomic data of mycoviruses, using semi-supervised learning.
- Presented the findings of this work at the ML+X forum held at the Wisconsin Institute for Discovery, presentation available [here](#).

Software Engineer Intern | *NimbleBox.ai* | Chennai, India | Jul 2022 - Aug 2023

- Fine-tuned several LLMs for a GenAI platform and Curated end-to-end API tests using Postman.
- Developed IAC templates on Pulumi to launch VMs and wrote DockerFiles to install, run and authenticate into Azure's CLI on VMs.
- Built an internal notification system, as part of the administrator panel of the Full-Stack MLOps Platform of NimbleBox.

PUBLICATIONS

- **Deep learning reveals functional archetypes in the adult human gut microbiome that underlie interindividual variability and confound disease signals.** Mohamed Meawad, Dalwinder Singh, Alice Deng, **Rohan Sonthalia**, Evelyn Cai, Vanessa Dumeaux. Available at: [Paper link](#).

TECHNICAL SKILLS

Programming Languages Python, Java, JavaScript, C, Golang, C++, R, Matlab, HTML/CSS
ML/AI LLMs, Transformers, LoRA, LangGraph, RAG, vLLM, NLP, PyTorch, TensorFlow, Keras
Infra/Tools Docker, Kubernetes, Azure, GCP, Bash, SQL, vectorDB, Postman, DataBricks, Terraform, Pulumi

PROJECTS

BioKlustering: Web App for phenotype predictions from genotype | *Python, Transformers, PyTorch* | Aug 2023 – Sept 2024

- Implemented 3 different methods of DNA sequence embedding: transformers, autoencoders and kmer-cants to vectors.
- Deployed relevant results onto an interactive web-app for real time conclusions from DNA sequence files.

Automated Online Video KYC Tool | *Python, OpenCV, ReactJS, gTTS* | Mar 2024 – Mar 2024

- Used OpenCV and gTTS libraries to automate KYC using Audio and Images for a simpler UX for wider Indian population.
- Built an interactive UI that could be displayed in several regional languages for the benefit of the user.